

Assignment 4 – Git and GitHub

Question 1

Objective

Create a GitHub repository, clone it using SSH, create a username branch, copy Assignment 3 Flask project, commit the changes and merge into main.

Steps Performed

1. Created GitHub repository flask_assignment4.
2. Cloned repository using SSH.
3. Created branch sid2812.
4. Copied Assignment 3 Flask project.
5. Added files to Git staging area.
6. Committed changes.
7. Merged branch into main.

Screenshots

Screenshot 1 – Branch Creation

```
PS C:\Users\91951> cd .\flask_assignment4
PS C:\Users\91951\flask_assignment4> git branch
* main
  sid2812
  sid2812_new
```

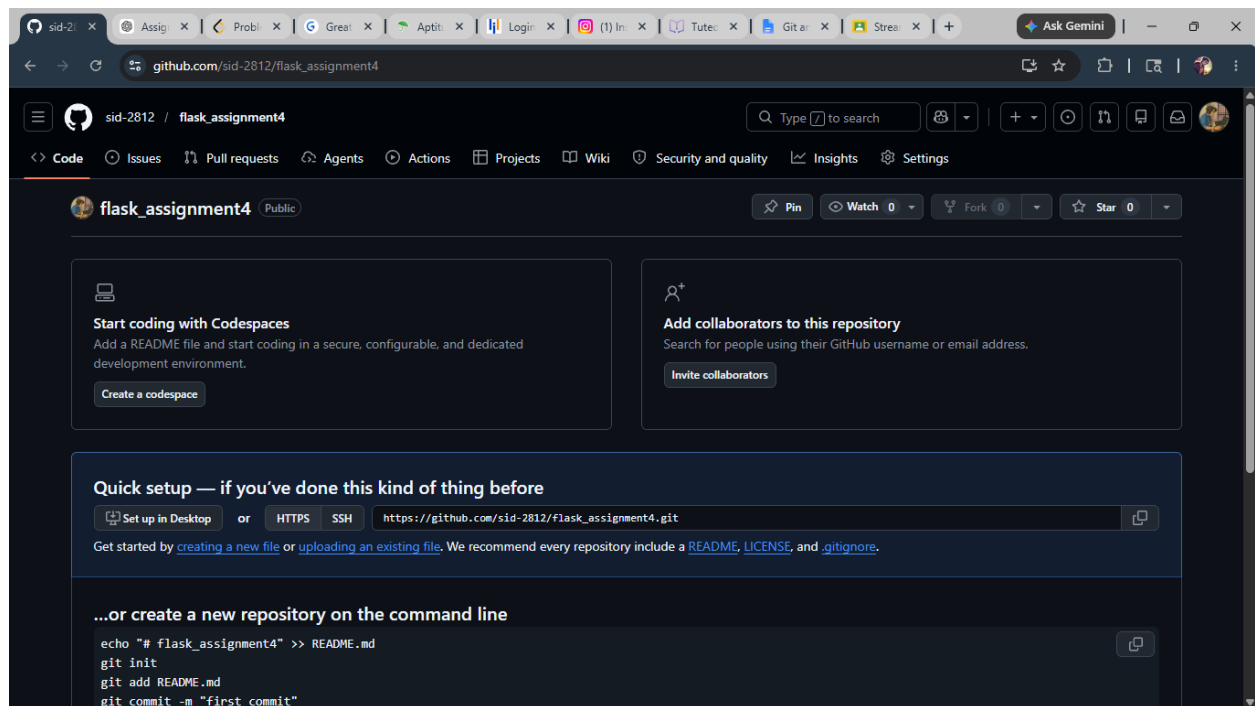
Screenshot 2 – Commit History

```
PS C:\Users\91951\flask_assignment4> git log --oneline --graph --all
* b993345 (HEAD -> main, origin/sid2812_new, origin/main, sid2812_new) Updated JSON data
* d52b8be (origin/sid2812, sid2812) Added flask project here from assignment 3
```

Screenshot 3 – Merge Result

```
PS C:\Users\91951\flask_assignment4> git checkout main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 3 commits.
(use "git push" to publish your local commits)
```

Screenshot 4 – GitHub Repository



Question 2

Objective

Modify the JSON file used by the Flask API route and commit the changes in a new branch.

Steps Performed

1. Created branch sid2812_new.
2. Modified q1/data.json.
3. Added updated data.
4. Committed changes.
5. Merged changes into main.

Screenshots

Screenshot 1 – Current Branches

```
PS C:\Users\91951> cd .\flask_assignment4
PS C:\Users\91951\flask_assignment4> git branch
* main
  sid2812
  sid2812_new
```

Screenshot 2 – Updated JSON Commit

```
PS C:\Users\91951\flask_assignment4> git log --oneline --graph --all
* b993345 (HEAD -> main, origin/sid2812_new, origin/main, sid2812_new) Updated JSON data
* d52b8be (origin/sid2812, sid2812) Added flask project here from assignment 3
```

Question 3

Objective

Create two branches:

- master_1 → To-Do page
- master_2 → MongoDB backend route

Part A – master_1

Steps Performed

1. Created branch master_1.
2. Created To-Do form.
3. Added Item Name field.
4. Added Item Description field.
5. Committed changes.

Screenshots

Screenshot 1 – master_1 Creation

```
PS C:\Users\91951\flask_assignment4> git checkout -b master_1
Switched to a new branch 'master_1'
PS C:\Users\91951\flask_assignment4> |
```

Screenshot 2 – Branch List

```
PS C:\Users\91951\flask_assignment4> git branch
main
* master_1
  sid2812
  sid2812_new
PS C:\Users\91951\flask_assignment4>
```

Screenshot 3 – q2 Structure

```
PS C:\Users\91951\flask_assignment4> dir q2

Directory:
C:\Users\91951\flask_assignment4\q2

Mode                LastWriteTime         Length
----                -
d-----          5/30/2026   9:05 AM
-a-----          12/4/2025   5:03 PM             854
-a-----          12/4/2025   4:46 PM              12
```

Screenshot 4 – Template Files

```
PS C:\Users\91951\flask_assignment4> Get-ChildItem q2\templates
```

Directory: C:\Users\91951\flask_assignment4\q2\templates

Mode	LastWriteTime	Length	Name
-a----	12/4/2025 4:43 PM	475	form.html
-a----	12/4/2025 4:44 PM	138	success.html
-a----	12/4/2025 4:44 PM	138	Untitled-1.html

Screenshot 5 – Modified Form

```
PS C:\Users\91951\flask_assignment4> git add .
PS C:\Users\91951\flask_assignment4> git commit -m "Added todo page"
[master_1 e847703] Added todo page
 1 file changed, 10 insertions(+), 12 deletions(-)
PS C:\Users\91951\flask_assignment4>
```

```
PS C:\Users\91951\flask_assignment4> git status
On branch master_1
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   q2/templates/form.html

no changes added to commit (use "git add" and/or "git commit -a")
PS C:\Users\91951\flask_assignment4> |
```

Screenshot 7 – Push master_1

```
PS C:\Users\91951\flask_assignment4> git push origin master_1
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 4 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 615 bytes | 123.00 KiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'master_1' on GitHub by visiting:
remote:   https://github.com/sid-2812/flask_assignment4/pull/new/master_1
remote:
To github.com:sid-2812/flask_assignment4.git
 * [new branch]      master_1 -> master_1
PS C:\Users\91951\flask_assignment4> |
```

Part B – master_2

Steps Performed

1. Created branch master_2.
2. Added /submittodoitem route.
3. Added MongoDB insertion logic.
4. Committed changes.
5. Pushed branch.

Screenshots

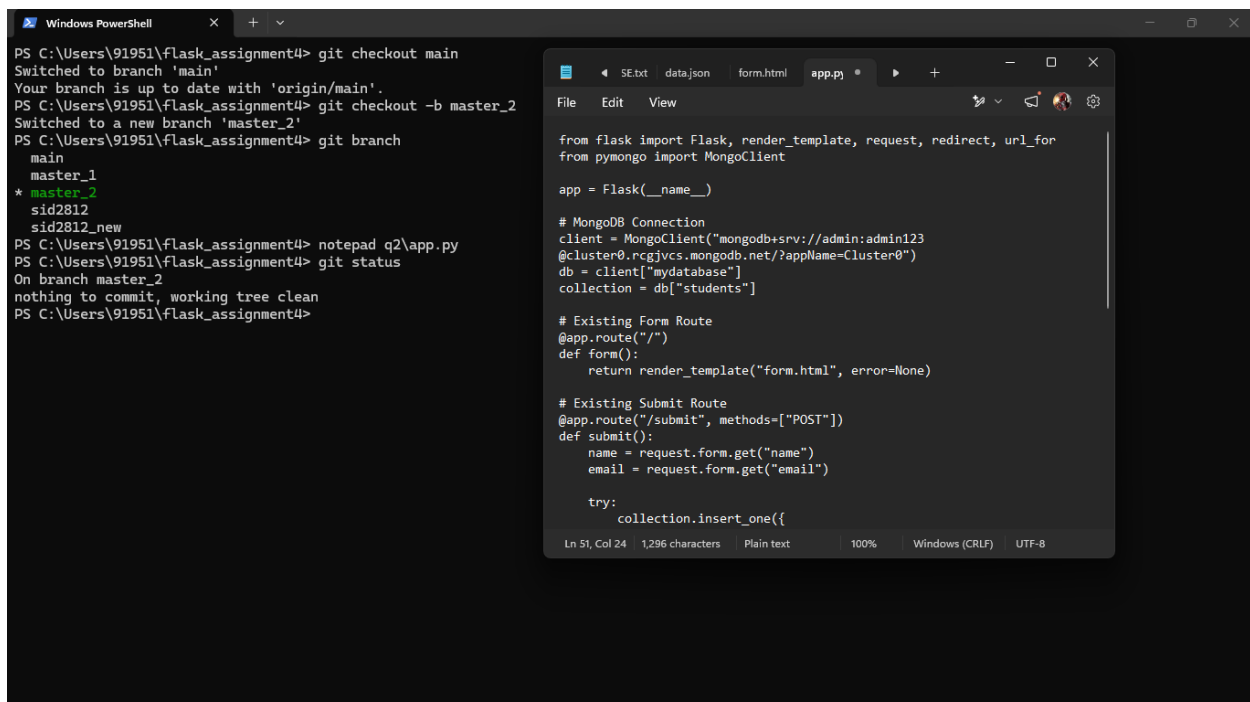
Screenshot 8 – master_2 Creation

```
PS C:\Users\91951\flask_assignment4> git checkout -b master_2
Switched to a new branch 'master_2'
PS C:\Users\91951\flask_assignment4> |
```

Screenshot 9 – Branch List

```
PS C:\Users\91951\flask_assignment4> git branch
main
master_1
* master_2
sid2812
sid2812_new
PS C:\Users\91951\flask_assignment4> |
```

Screenshot 10 – Backend Route Modified



The screenshot shows a Windows PowerShell window on the left and a code editor on the right. The PowerShell window displays the following commands and output:

```
PS C:\Users\91951\flask_assignment4> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS C:\Users\91951\flask_assignment4> git checkout -b master_2
Switched to a new branch 'master_2'
PS C:\Users\91951\flask_assignment4> git branch
main
master_1
* master_2
sid2812
sid2812_new
PS C:\Users\91951\flask_assignment4> notepad q2\app.py
PS C:\Users\91951\flask_assignment4> git status
On branch master_2
nothing to commit, working tree clean
PS C:\Users\91951\flask_assignment4>
```

The code editor on the right shows the contents of the file `app.py`. The code defines a Flask application with a MongoDB connection and two routes: a form route and a submit route.

```
from flask import Flask, render_template, request, redirect, url_for
from pymongo import MongoClient

app = Flask(__name__)

# MongoDB Connection
client = MongoClient("mongodb+srv://admin:admin123@cluster0.rcgjcvs.mongodb.net/?appName=Cluster0")
db = client["mydatabase"]
collection = db["students"]

# Existing Form Route
@app.route("/")
def form():
    return render_template("form.html", error=None)

# Existing Submit Route
@app.route("/submit", methods=["POST"])
def submit():
    name = request.form.get("name")
    email = request.form.get("email")

    try:
        collection.insert_one({
```

Screenshot 11 – Backend Commit

```
nothing to commit, working tree clean
PS C:\Users\91951\flask_assignment4> git add .
PS C:\Users\91951\flask_assignment4> git commit -m "Added submit todo item route"
On branch master_2
nothing to commit, working tree clean
PS C:\Users\91951\flask_assignment4> |
```

Screenshot 12 – Push master_2

```
nothing to commit, working tree clean
PS C:\Users\91951\flask_assignment4> git push origin master_2
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'master_2' on GitHub by visiting:
remote:   https://github.com/sid-2812/flask_assignment4/pull/new/master_2
remote:
To github.com:sid-2812/flask_assignment4.git
 * [new branch]      master_2 -> master_2
PS C:\Users\91951\flask_assignment4> |
```

Screenshot 13 – Final Git History

```
PS C:\Users\91951\flask_assignment4> git log --oneline --graph --all --decorate
*   df651a7 (HEAD -> main) Merged master_1 into main
|  \
|   * ac56ef0 (origin/master_2, master_2) Added submittodoitem route
|  * | e847703 (origin/master_1, master_1) Added todo page
| /
*   b993345 (origin/sid2812_new, origin/main, sid2812_new) Updated JSON data
*   d52b8be (origin/sid2812, sid2812) Added flask project here from assignment 3
PS C:\Users\91951\flask_assignment4> |
```

Question 4

Objective

Perform multiple commits, reset operation and rebase operation.

Commit 1

Added:

Item ID

Screenshot

```
PS C:\Users\91951\flask_assignment4> git add .
PS C:\Users\91951\flask_assignment4> git commit -m "Added Item ID field"

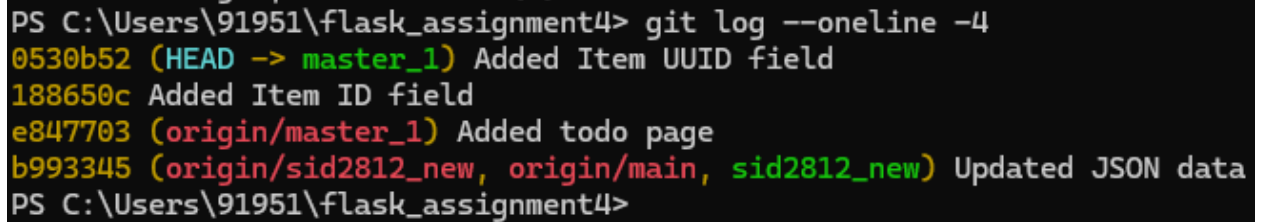
[master_1 188650c] Added Item ID field
 1 file changed, 2 insertions(+), 1 deletion(-)
PS C:\Users\91951\flask_assignment4>
```

Commit 2

Added:

Item UUID

Screenshot



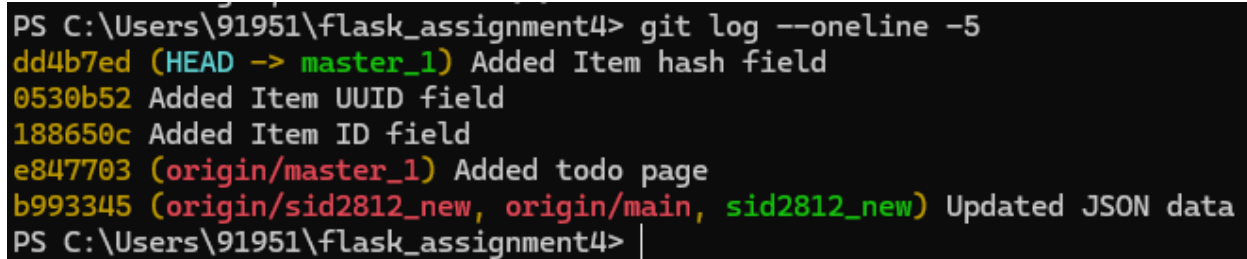
```
PS C:\Users\91951\flask_assignment4> git log --oneline -4
0530b52 (HEAD -> master_1) Added Item UUID field
188650c Added Item ID field
e847703 (origin/master_1) Added todo page
b993345 (origin/sid2812_new, origin/main, sid2812_new) Updated JSON data
PS C:\Users\91951\flask_assignment4>
```

Commit 3

Added:

Item Hash

Screenshot



```
PS C:\Users\91951\flask_assignment4> git log --oneline -5
dd4b7ed (HEAD -> master_1) Added Item hash field
0530b52 Added Item UUID field
188650c Added Item ID field
e847703 (origin/master_1) Added todo page
b993345 (origin/sid2812_new, origin/main, sid2812_new) Updated JSON data
PS C:\Users\91951\flask_assignment4> |
```

Soft Reset

Command Used:

git reset --soft 188650c

Screenshot

```
PS C:\Users\91951\flask_assignment4> git reset --soft
PS C:\Users\91951\flask_assignment4> git reset --soft 188650c
PS C:\Users\91951\flask_assignment4> |
```

Recommit

Created new commit after reset.

Screenshot

```
PS C:\Users\91951\flask_assignment4> git log --oneline -6
fb9d9e8 (HEAD -> main) Reset state to Item ID
188650c Added Item ID field
e847703 Added todo page
b993345 (origin/sid2812_new, origin/main, sid2812_new) Updated JSON data
d52b8be (origin/sid2812, sid2812) Added flask project here from assignme
nt 3
PS C:\Users\91951\flask_assignment4> |
```

Rebase

Performed:

git rebase main

Conflict occurred in:

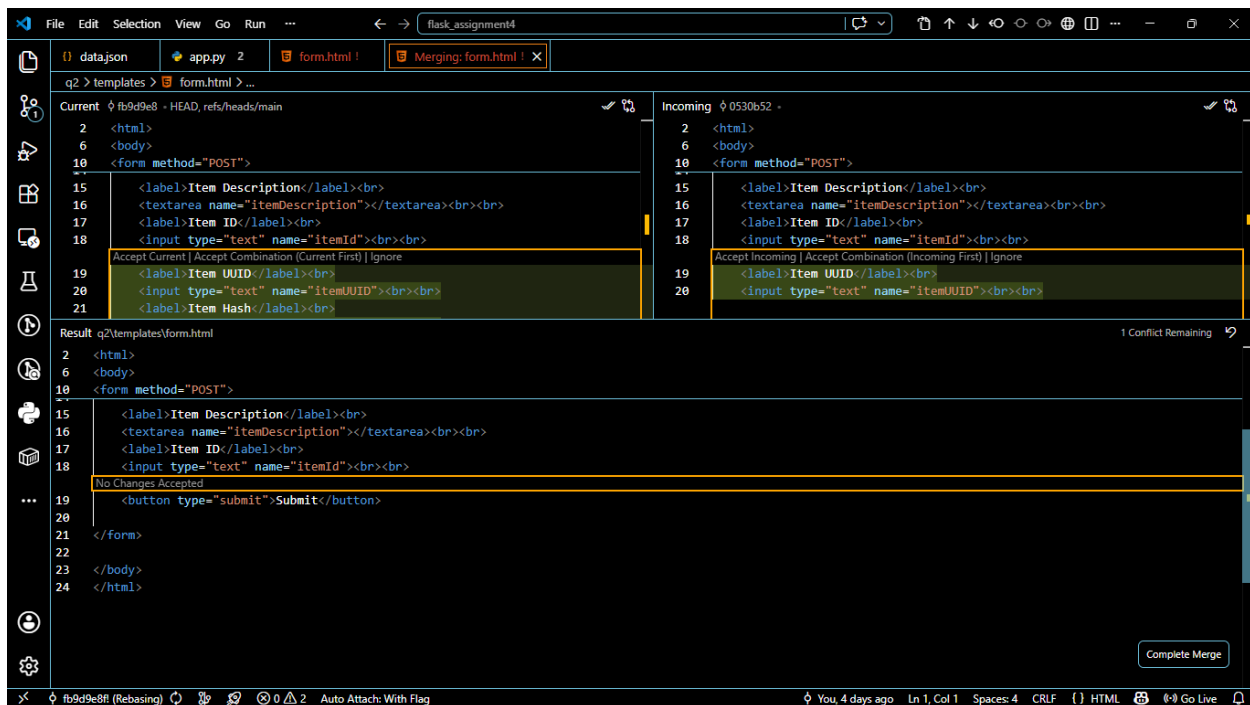
q2/templates/form.html

Conflict was resolved manually and rebase completed.

Screenshots

Rebase Conflict

Paste:



Final Git History

```
PS C:\Users\91951\flask_assignment4> git status
On branch master_1
nothing to commit, working tree clean
PS C:\Users\91951\flask_assignment4> git log --oneline --graph --all --decorate -10
* fb9d9e8 (HEAD -> master_1, main) Reset state to Item ID
| * dd4b7ed (origin/master_1) Added Item hash field
| * 0530b52 Added Item UUID field
|/
* 188650c Added Item ID field
* e847703 Added todo page
| * ac56ef0 (origin/master_2, master_2) Added submittodoitem route
|/
* b993345 (origin/sid2812_new, origin/main, sid2812_new) Updated JSON data
* d52b8be (origin/sid2812, sid2812) Added flask project here from assignment 3
PS C:\Users\91951\flask_assignment4> |
```

Conclusion

Successfully performed Git and GitHub operations including repository creation, cloning using SSH, branching, commits, merging, JSON updates, Flask application modifications, MongoDB integration, reset operation and rebase operation.